

Build a color-test station

Name: _____ Team: _____ Date: _____

A wheeled robot base and a set of cards that make the screen respond.

Gather these materials

- [] LEGO set #45522 and its official instructions
- [] Connected computer or tablet
- [] Red, white and blue paper: one 8 cm square of each
- [] Two equal-height firm supports, checked by the teacher

Build: steps 1-4

- [] 1. Open LEGO's official instruction book. Find model C103 on pages 24-31. Build it with your teacher as a starting model, or use the wheeled base your teacher has already checked. The pictures in this course are project ideas, not exact LEGO assembly steps.
- [] 2. Ask the teacher to prepare a secure downward-facing color-sensor mount and clear attachment points for later lessons. Keep both wheel paths and all Stop controls clear. This is a change to the starting model and needs a real-kit test.
- [] 3. Set the robot on the two firm supports under fixed parts of its frame. Check that it cannot rock or slide. Both wheels must hang clear of the table. Do not support it by the sensor or an axle (a rod a wheel turns on).
- [] 4. Cut three paper squares, each 8 cm by 8 cm. Use red, white and blue. Put them beside the robot. Keep the same side of each paper facing the sensor.

[Open official LEGO instructions: C103, pages 24-31](#)

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

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Build: steps 5-8

[] 5. Connect only your team's motor and sensor in Coding Canvas. Point to the screen Stop control and the robot's power button. Ask the teacher to check the connection.

[] 6. Slide white paper below the sensor without touching it. Adjust the paper position until the sensor reads white. Mark that place with tape so every card goes in the same spot.

[] 7. Build the color-to-message program below. Keep all movement stopped. Test one card at a time, then put it back in the marked spot.

[] 8. Record all three results. Change one screen message to your team's words and test again. Leave the sensor mount and base ready for lesson 2.

Make the program

Start with all movement stopped.

Repeat 10 times: read the color. White shows READY; red shows DELIVERY; anything else shows CHECK. Wait 0.2 seconds before the next check.

Stop all movement and end. There are no driving steps in this program.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your color readings

Card color	Expected message	Actual message	Wheels still?
White			
Red			
Blue			

A successful build

The base is steady and both wheels stay still.

Each color produces the message in the rule.

Everyone can point to the sensor and the Stop controls.

How does the color make the screen message change?